Data-500 Project Presentation

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Objective

Our objective for this report was to identify patient attributes that are most indicative of Alzheimer's disease, with the goal of helping flag patients who may be at risk and offering early intervention.

Raw Data

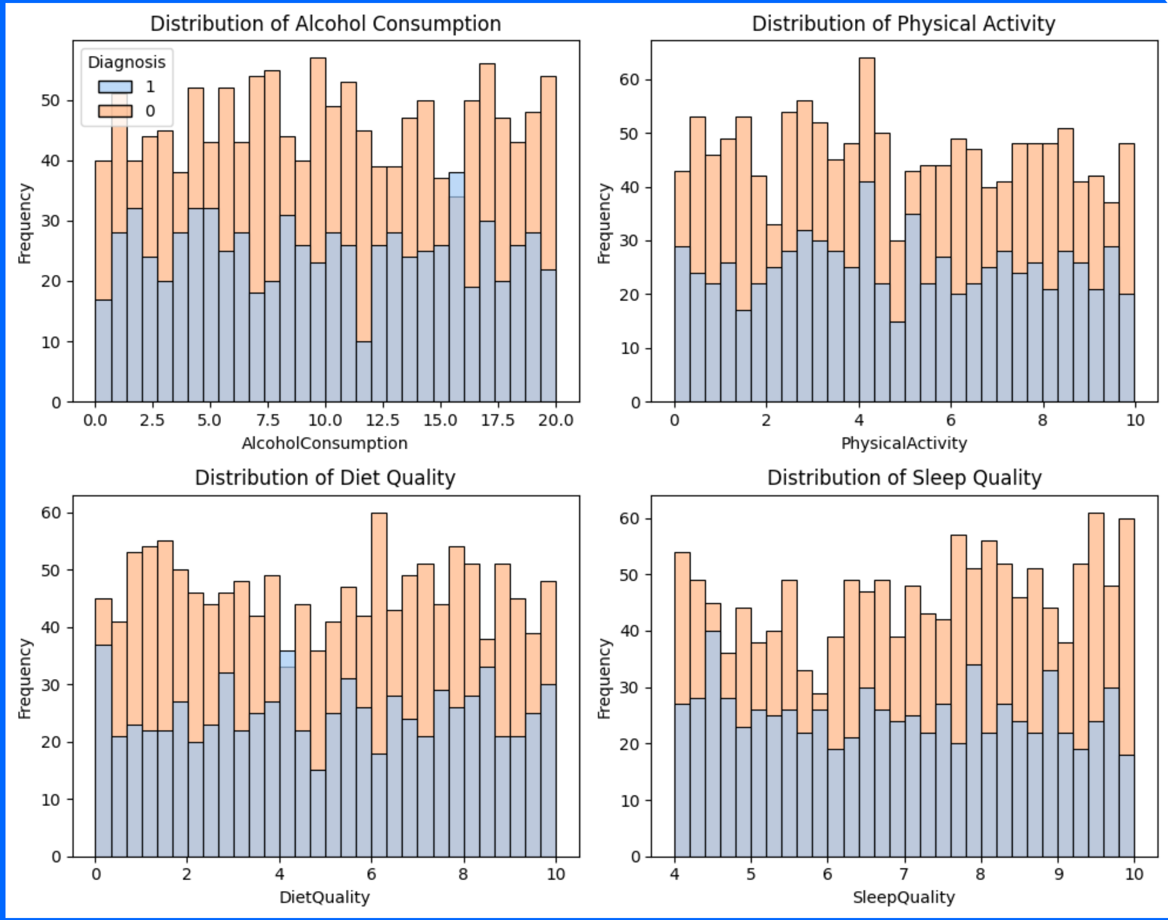
Column name	Type	Description
PatientID	Numeric	Unique identifier assigned to each patient (4751-6900)
Age	Numeric	The age of the patients (60-90)
Gender	Numeric	0-male, 1-Female
Ethnicity	Numeric	0-Caucasian, 1-African American, 2-Asian, 3-Other
EducationLevel	Numeric	0-None, 1-High School, 2-Bachelor’s, 3-Higher
BMI	Numeric	Body mass of the patients (15-40)
Smoking	Numeric	0-No, 1-Yes
AlcoholConsumption	Numeric	Weekly consumption in units (0-20)
PhisicalActivity	Numeric	Weekly physical activity in hours (0-10)
DietQuality	Numeric	Diet quality score (0-10)
SleepQuality	Numeric	Sleep quality score (4-10)
FamilyHistoryAlzheimers	Numeric	0-Family history, 1-No History
CardiovascularDisease	Numeric	0-No, 1-Yes
Diabetes	Numeric	0-No, 1-Yes
Depression	Numeric	0-No, 1-Yes
HeadInjury	Numeric	0-No, 1-Yes
Hypertension	Numeric	0-No, 1-Yes
SystolicBP	Numeric	Systolic blood pressure (90-180) mmHg

Column name	Type	Description
DiastolicBP	Numeric	Diastolic blood pressure (60-120) mmHg
CholesterolTotal	Numeric	Total cholesterol levels (150-300)mg/dL
CholesterolLDL	Numeric	Low-density lipoprotein cholesterol levels (50-200) mg/dL
CholesterolHDL	Numeric	High-density lipoprotein cholesterol levels (20-100) mg/dL
CholesterolTriglycerides	Numeric	Triglycerides levels (50-400) mg/dL
MMSE	Numeric	Mini-Mental State Examination score (0-30). Lower scores indicate cognitive impairment.
FunctionalAssessment	Numeric	Functional assessment score (0-10). Lower scores indicate greater impairment.
MemoryComplaints	Numeric	0-No, 1-Yes
BehavioralProblems	Numeric	0-No, 1-Yes
ADL	Numeric	Activities of Daily Living score (0-10). Lower scores indicate greater impairment.
Confusion	Numeric	0-No, 1-Yes
Disorientation	Numeric	0-No, 1-Yes
PersonalityChanges	Numeric	0-No, 1-Yes
DifficultyCompletingTasks	Numeric	0-No, 1-Yes
Forgetfulness	Numeric	0-No, 1-Yes
Diagnosis	Numeric	0-No Alzheimer’s (65%), 1-Alzheimer’s (35%)
DoctorInCharge	Character	Contains confidential information about the doctor in charge, "XXXConfid" is the value for all patients.

Exploratory Data Analysis

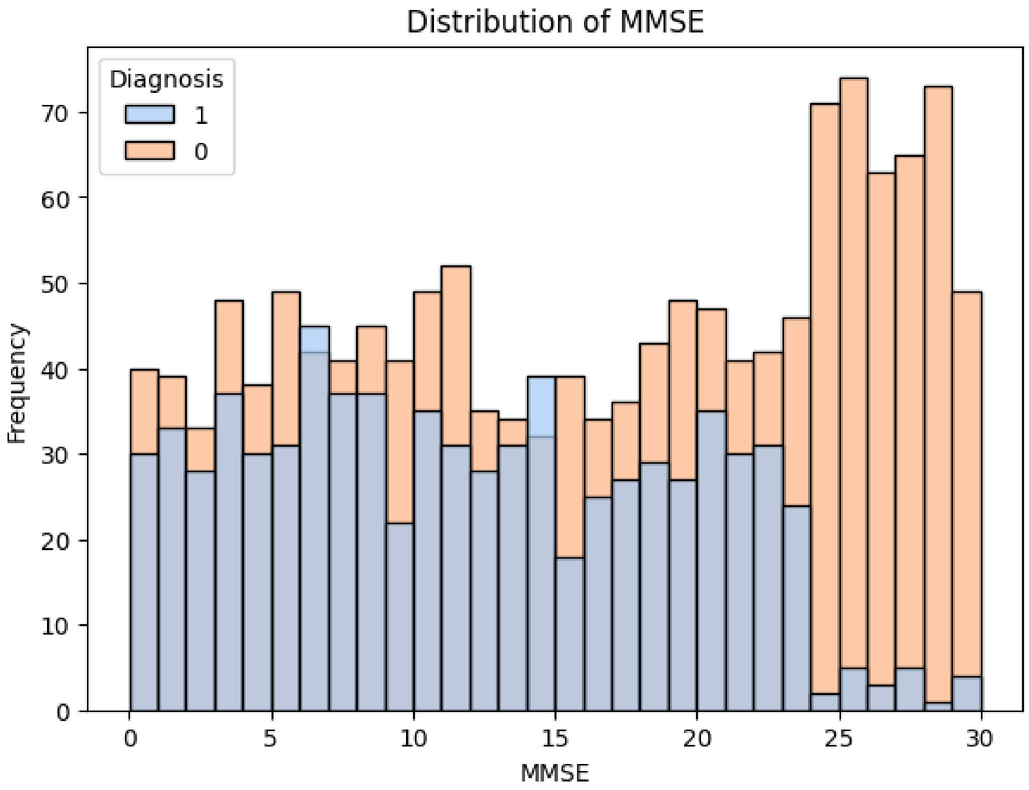
Lifestyle Habits Distribution:

- All lifestyle variables show similar distribution shapes.
- There is a similar frequency of diagnosis across the entire range of each variable.



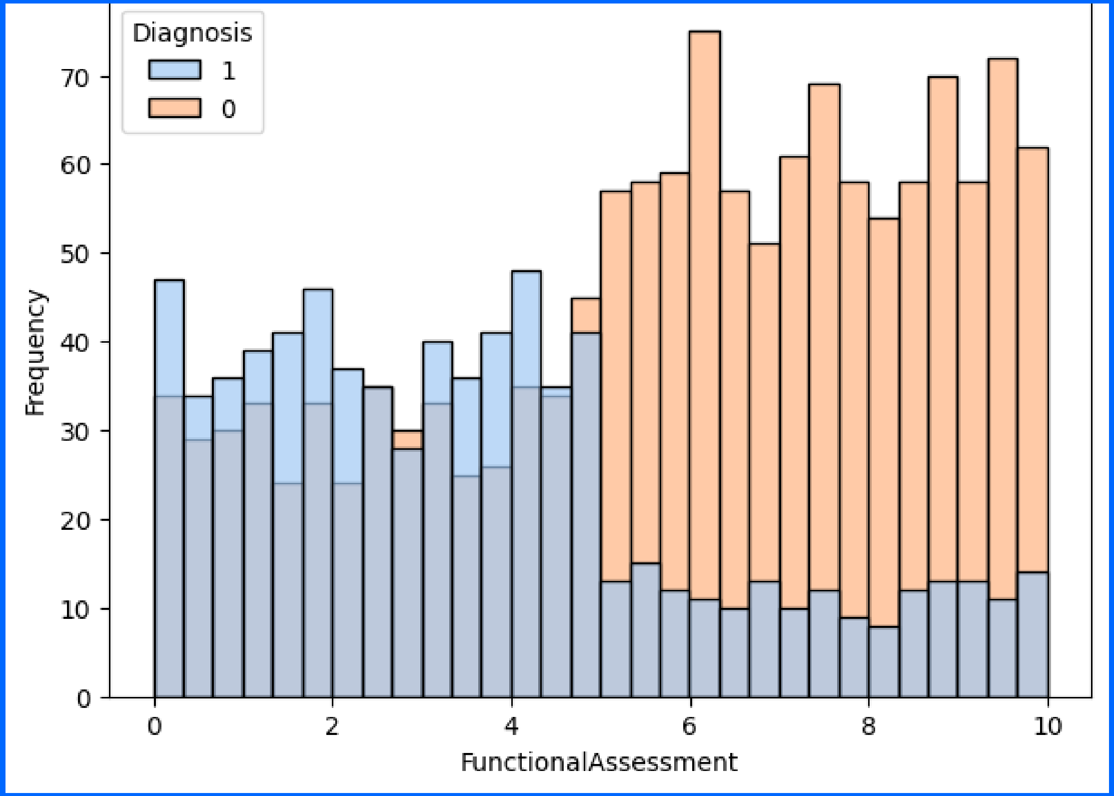
Mini-Mental State Examination (MMSE) Score Distribution:

- Very few Alzheimer's diagnosed patients scored above a 25.
- Lower scores indicate cognitive impairment, which explains why fewer positively diagnosed patients scored high on the MMSE.



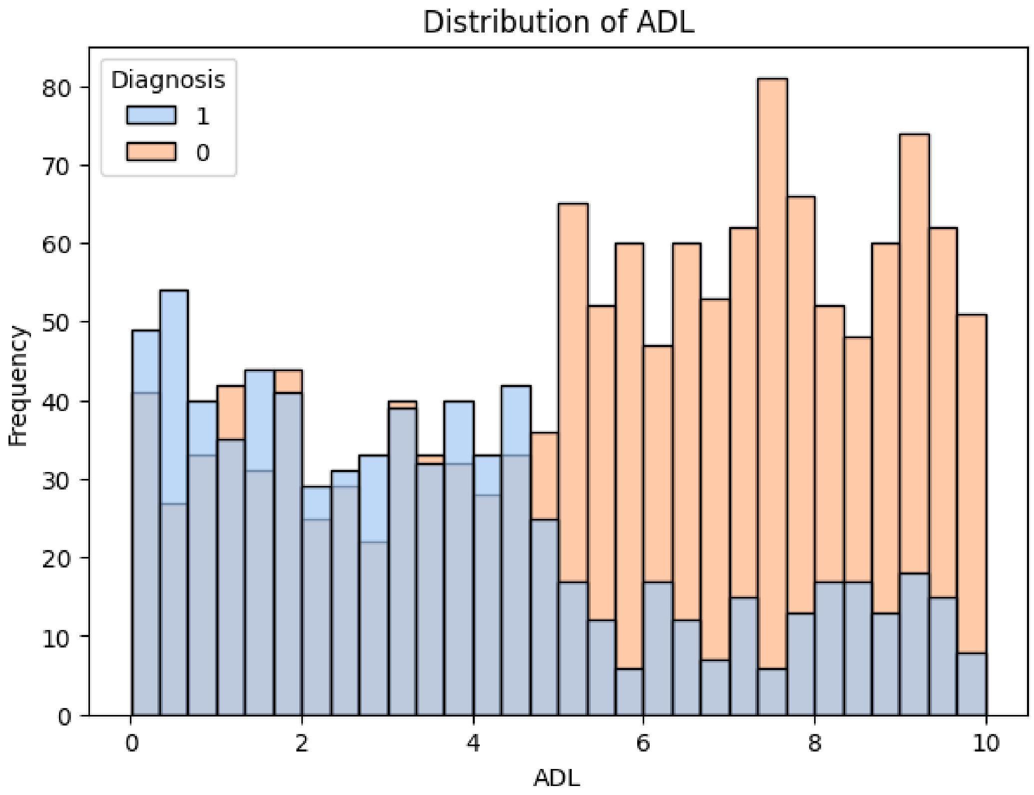
Functional Assessment Score Distribution:

- The distribution of positive cases (Alzheimer's diagnosis) is right-skewed, indicating greater impairment in these patients.
- The distribution of negative cases (non-diagnosed patients) is left-skewed, suggesting less impairment.



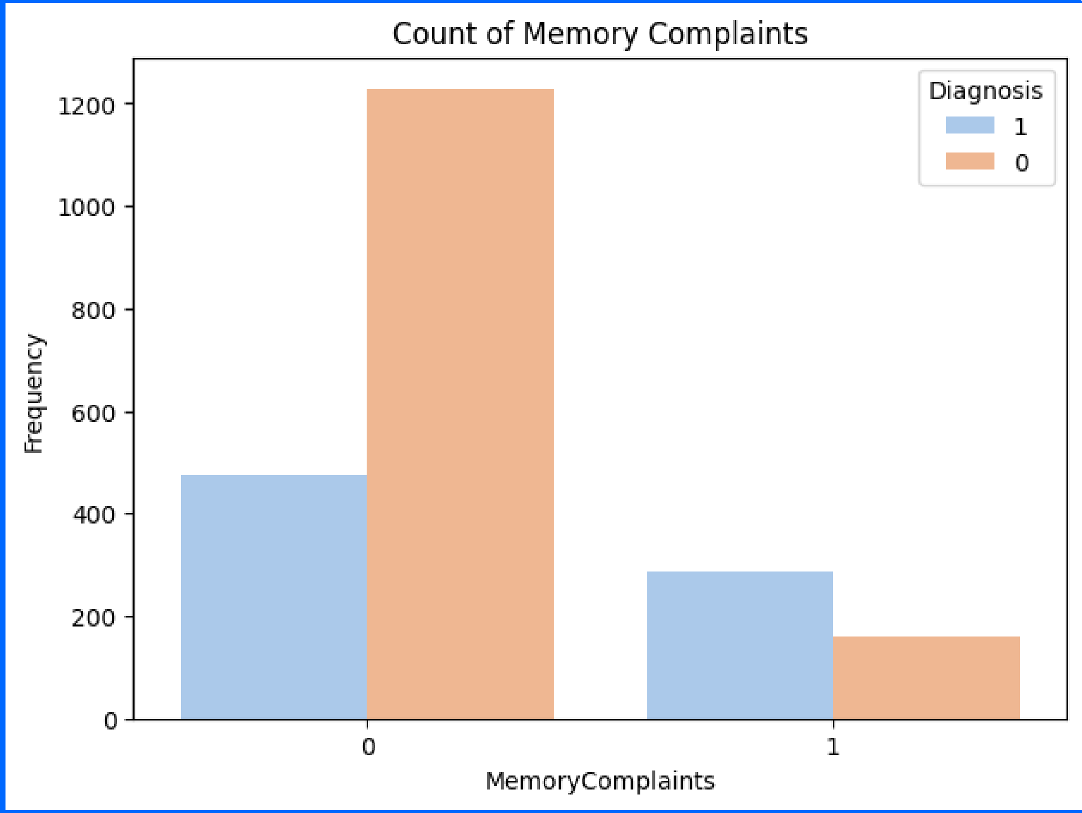
Activities of Daily Living (ADL) Score Distribution:

- The distribution of positive cases (Alzheimer's diagnosis) is right-skewed, indicating greater impairment in these patients.
- The distribution of negative cases (non-diagnosed patients) is left-skewed, suggesting less impairment.



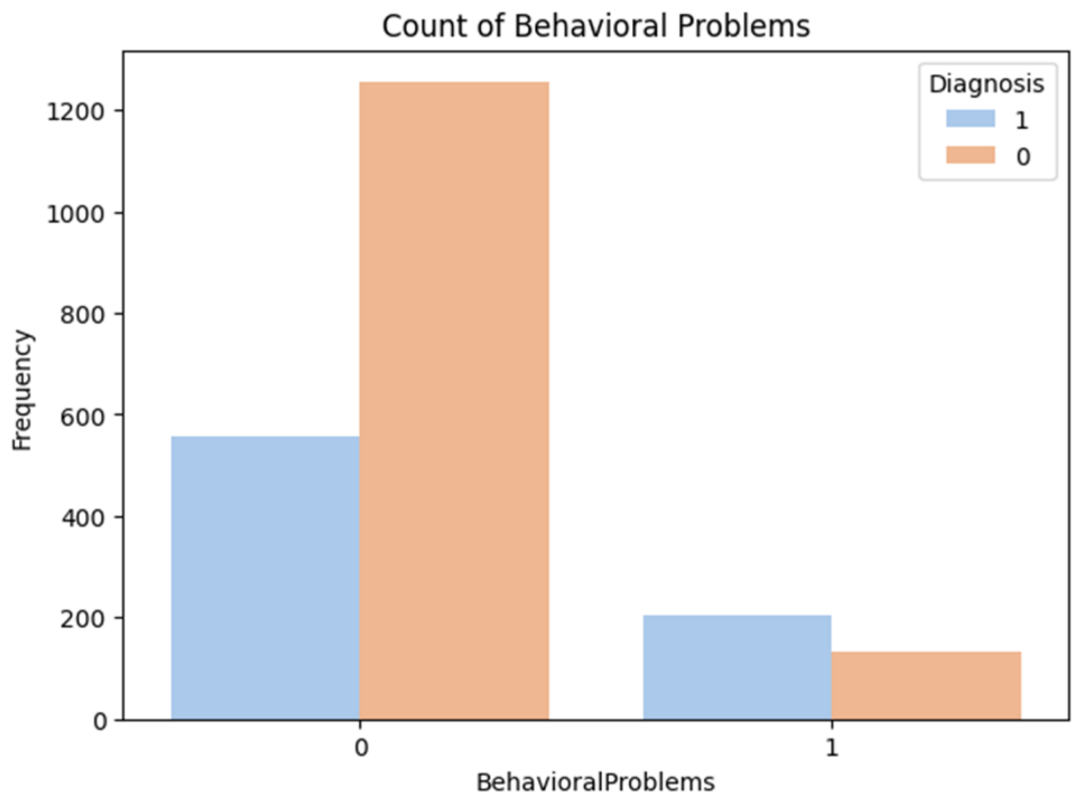
Memory Complaints Countplot:

- More patients diagnosed with Alzheimer’s reported memory complaints than patients without.



Behavioral Problems Countplot:

- More patients diagnosed with Alzheimer’s reported behavioral problems than patients without.



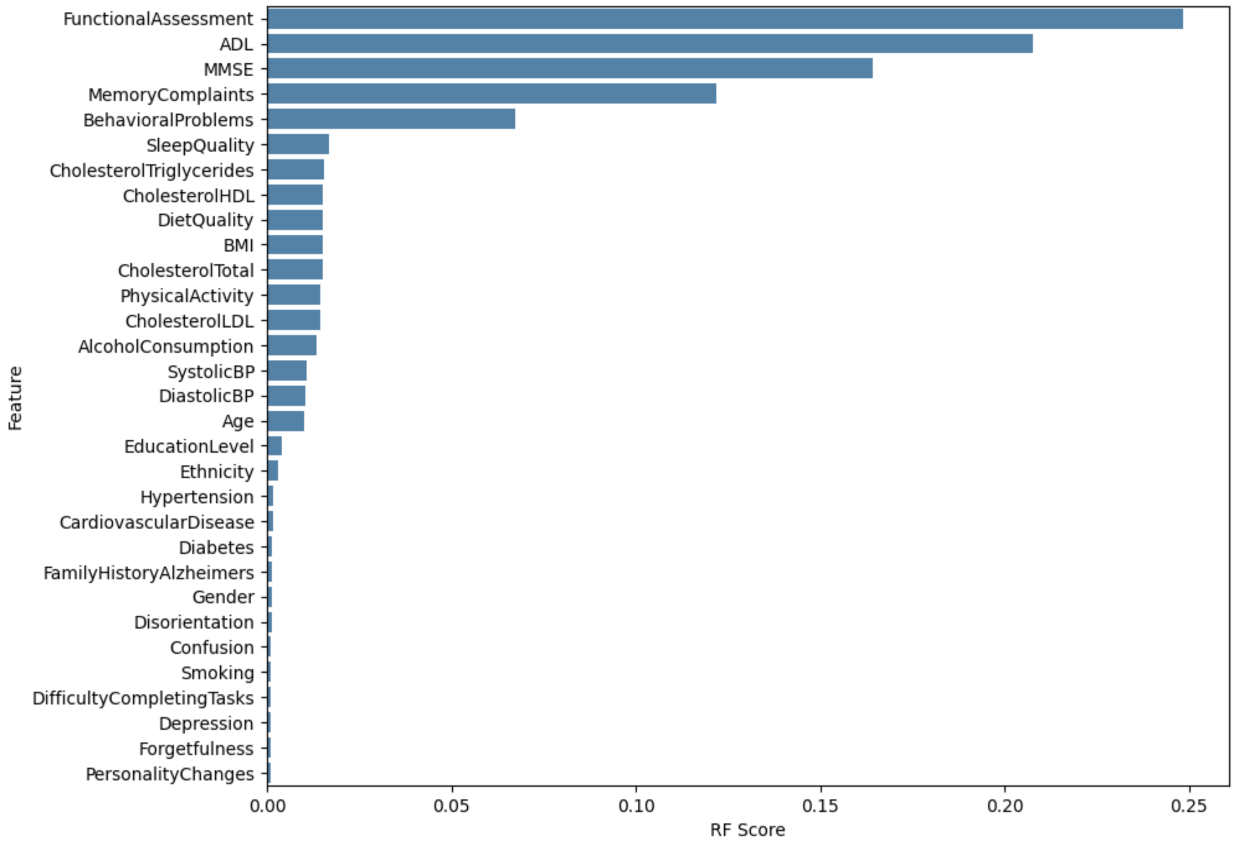
Modeling

Random Forest Feature Importances:

- Chart shows feature importances ordered by their impact on the model.
- The top 5 important features align with the variables identified in the exploratory analysis.
- This confirms that our initial assumptions about impactful features for Alzheimer's diagnosis were accurate.

Model Performance:

- Achieved 89% accuracy using the Random Forest model with 5-fold cross-validation for predicting Alzheimer's diagnosis.



Conclusion

Objective: Identify key patient attributes for early Alzheimer's intervention and risk flagging.

EDA Findings:

- No significant correlation between lifestyle factors (alcohol, physical activity, sleep, diet) and Alzheimer's diagnosis.
- Similar distributions for these factors in both Alzheimer's positive and negative patients.
- These lifestyle factors are not strong predictors of Alzheimer's.

Key Indicators for Alzheimer's Diagnosis:

- Memory complaints and behavioral problems were significant indicators.
- Mental and physical exams were the most important for predicting diagnosis.
- Early identification of these attributes could lead to faster diagnoses and better patient outcomes.

Further Study

Future Directions:

- Explore interactions between variables to identify potential new features.
- Apply feature engineering techniques to create effective predictors for Alzheimer's.

Further Model Exploration:

- Achieved high accuracy with a simple random forest model.
- Investigate more complex models to see if we can achieve higher accuracy.

Works Cited

Dataset:

https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset/data[1]

Alzheimer's background information:

https://www.alz.org/alzheimer_s_dementia[2]